MILITARY SPECIFICATION

ELECTRONICS ASSEMBLY, FOR MINES, M718/M741, SYSTEM

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by drawings and provides Quality Assurance provisions for the fabrication of parts, assembly and packing of the electronic assembly used in the M718/M741, 155MM Projectile, AT/AV Mine.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications For
MIL-A-63242	-	Amplifier, Operation, Triple Programmable Micropower
MIL-S-45743	-	Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment

FSC 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-46844 - Solder Bath Soldering of Printed Wiring Assemblies

MIL-S-19500 - Semiconductor Devices General Specification For

MIL-R-39008 - Resistors, Fixed, Composition (Insulated)

MIL-R-39008/1 - Resistors, Fixed, Composition (Insulated) Established Reliability Style RCR07

MIL-P-63238 - Projectile, 155MM, AT M718/M741 Loading, Assembling and Packing Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributers

MIL-STD-202 - Test Methods For Electronic and Electrical Component Parts

MIL-STD-750 - Test Methods For Semiconductor Devices

MIL-STD-883 - Test Methods and Procedures For Microelectronics

MIL-STD-1169 - Packaging, Packing and Marking For Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent herein.

DRAWINGS

U. S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9287150-1, -2	Electronic Assembly (M718)
9287151-1, -2	Electronic Assembly (M741)
9287153	Performance Requirements
9317597	Projectile, 155MM Test, Non-Haz, M718 Type
9317598	Projectile, 155MM Test, Non-Haz, M741 Type

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D2240-68 - Indentation Hardness of Rubber and Plastic by Means of a Durometer, Test For

ASTM D412-68 - Tension Testing of Vulcanized Rubber

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA., 19102).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9287150-1, -2, and 9287151-1, -2, and 9287153, and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Physical characteristics.

3.3.1 Envelope and dimensions. The envelope and dimensions of the electronic assembly shall conform to the requirements on drawings 9287150 or 9287151 as applicable.

3.3.2 Potting. Potting material shall comply with the requirements of drawing 9287115 and 9287148 and shall be manufactured and dispensed in accordance with the processes, procedures and controls delineated therein. The potted electronic assembly shall not show evidence of voids and/or bubbles exceeding the following criteria:

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a) Bubbles that are less than 1/4 inch in the maximum physical dimension are permissible. The total volume occupied by bubbles or voids cannot exceed 1/8 cu. in. (1/2 X 1/2 X 1/2" cube). In addition, no bubbles shall be in contact with any electronic component except that the power supply is allowed to trap bubbles under the lip of the battery on the contact end. (A bubble is defined to be entrapped air with no exposure to the surface.)

b) Single or multiple voids less than 1/4 inch in the maximum physical dimension are permissible on the surface except as defined in c and d. The total volume occupied by bubbles or voids cannot exceed 1/8 cu. in. (1/2 X 1/2 X 1/2" cube) and must not expose any components or the printed circuit board. (A void is defined as an irregularity of any shape which is open to the surface.)

c) Voids on the outer circumference of the lens are permissible provided no component or part of the printed wiring board is exposed.

d) Voids on the flat vertical wall of the "D Hole" (and associated channels) up to a maximum of 25% of the depth will be allowed.

e) Sprue cutoff deviations shall be allowed to +.000" and -.040" of the curved surface.

f) Slight irregularities on the spherical surface (associated with the potting prematurely separating from the mold) within a 1" radius of the "Sprue" center are permissible.

g) A potting void or hole at the bottom of the "D" hole, exposing the battery case, of a maximum size of 1/8 inch long by 1/16 inch wide or 3/32 inch diameter is allowable provided the battery case is below bottom surface of the "D" hole. Larger voids will have potting material applied to cover the battery case and allowed to air dry.

Delamination of potting from the printed wiring board shall not be allowed (non-destructively viewed from the top of the board only). Components and item designation (M718/M741 shall be clearly visible through the potting material.

3.4 Environmental conditions.

3.4.1 Temperature. The electronic assembly shall perform as required herein and in drawing 9287153 over a temperature range from -25°F to +145°F.

3.4.2 Ballistic test. The electronic assembly shall perform as required after exposure to gun launch and ground impact environments of zone 8 in the M109A1 gun at 145°F. (See 4.5.25)

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3.5 Electrical requirements.

3.5.1 Quiescent current. Current drawing shall be less than 500 microamperes when powered by a supply voltage (VB) of 6.75 volts.

3.5.2 Self-destruct timing control and firing circuit characteristics. The electronic assembly shall provide timing control and firing signals as specified in drawing 9287153 and the following paragraphs. Unless otherwise specified VB = 6.75 volts and ambient temperature (TA) shall be 75°F.

3.5.2.1 Oscillator function. The primary time base (PTB) oscillator and the test time base (TTB) oscillator circuits shall oscillate at the periods specified in drawing 9287153 over the temperatures specified. (See 4.5.2, 4.5.3)

3.5.2.2 Arming delay/sensor enable. The electronic assembly shall not propagate a fire signal from the sensor prior to the arming time specified on drawing 9287153. A total of 176 primary time base pulses shall have been generated after initial application of VB. (See 4.5.4 and 4.5.5)

3.5.2.3 Low voltage detection. The electronic assembly shall provide a detonator fire signal (see para. 3.5.2.7) after the above arming delay when the supply voltage is reduced to the levels specified in drawing 9287153 at the temperatures specified. (See 4.5.6).

3.5.2.4 Self-destruct.

3.5.2.4.1 Part number 9287150. The electronic assembly, P/N 9287150, shall provide a detonator fire signal (see para. 3.5.2.7) at the self-destruct time specified in drawing 9287153 after initial application of voltage (VB). A total of 392,448 primary time base pulses shall have been generated at the period specified in the drawing to meet the requirement. (See 4.5.7).

3.5.2.4.2 Part number 9287151. The electronic assembly shall provide a detonator fire signal (see para. 3.5.2.7) at the self-destruct time specified in drawing 9287153 after initial application of voltage (VB). A total of 32000 primary time base pulses shall have been generated at the period specified in the drawing to meet this requirement (see 4.5.8).

3.5.2.5 Oscillator check function. The electronic assembly shall provide a detonator fire signal (see para. 3.5.2.7), after the arming delay, when exactly two (2) periods of the test time base oscillator occur upon termination of the primary time base oscillations (see 4.5.9).

3.5.2.6 Disturbance. The electronic assemblies that contain anti-disturbance switches shall provide a detonator fire signal (see para. 3.5.2.7), after the arming delay, when the assembly is jarred, tilted, rotated or otherwise tampered with (see 4.5.10).

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3.5.2.7 Detonator fire signal output energy. The electronic assembly shall provide energy to fire the detonators when required as described in the specification. The output energy shall be such that a minimum of three (3) volts shall be measured no more than forty (40) microseconds after a fire output signal is generated and a minimum of two (2) volts is measured at a minimum time of 120 microseconds after the fire output signal is generated. Voltages shall be measured at the output terminals across a 3.9 ohm ± 5% resistive load (see 4.5.11).

3.5.2.8 Self-sterilization. Within 5 seconds after the turn on of the detonator firing circuit the voltage on capacitor C12 shall be less than 1.3 volts. It shall remain less than 1.3 volts for as long as the detonator firing circuit remains on (see 4.5.12).

3.5.3 Target sensor functioning. The electronics assembly shall provide a fire signal (para. 3.5.3.2) when subjected to valid target signatures. The specific conditions and the signal limit under which this function shall occur are specified in drawing 9287153 and the following paragraphs.

3.5.3.1 Spurious output signal. There shall be no output occurring at the sensor output test point (test point 1) for a minimum of ten (10) second prior to electronic arming after initial application of VB (see 4.5.13).

3.5.3.2 Sensor output voltage. The output from the target sensor (test point 1) shall be a minimum of 2.5 volts when the assembly is subjected to a valid signature as specified in drawing 9287153 (see 4.5.13).

3.5.4 Non-activated voltage. Each battery cell shall be unactivated and measured less than .005 volts when measured by a 10^7 ohm input impedance meter (see 4.5.14).

3.5.5 Soldering. All soldered electrical connections shall comply with MIL-S-46844, MIL-S-45743 and Appendix to this specification.

3.5.5.1 Solderability. Electronic and electric assembly parts that are to be soldered shall comply with the solderability requirements of MIL-S-45743 and MIL-S-46844 except integrated circuits shall be tested for solderability using the methods and acceptance criteria of MIL-STD-883 and diode and transistors shall be tested for solderability using the methods and acceptance criteria of MIL-STD-750 provided the leads are not gold plated. When gold plated leads are used, the provisions of MIL-S-45743 apply.

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3.5.5.2 Quality assurance soldering plan. The contractor will generate a Quality Assurance Soldering Plan that will detail the overall soldering procedure. This plan shall include detailed information for handling, cleaning, hand soldering, machine soldering, process controls including documentation required by 3.4.4 of MIL-S-46844 tool maintenance, flagging procedures for avoiding excessive defects and all other pertinent QA procedures relative to soldering. This plan will be generated and submitted to ARRADCOM, DRDAR-QAT-M (see 4.5.22).

3.5.6 Shorting bar continuity. There shall be no more than one (1) ohm resistance in the shorting bar loop when measured between the output terminal to ground (see 4.5.16).

3.6 Workmanship. The electronic assemblies shall be fabricated and finished in a thorough workmanlike manner. As a minimum, there shall be no potting material on the cover flange or in the battery primer hole. Sprue holes shall be ground to flush or below flush. All marking shall be neat and sharply defined. The cover shall be devoid of dirt, grease or other foreign matter. The surface coating shall be continuous except for a few light scratches not exposing base metal. All parts shall be free from chips, dirt, grease and other foreign matter.

3.7 Inspection data. Data shall be recorded and retained for all First Article and Quality Conformance tests required by this specification in accordance with 4.5.21.

3.7.1 Variables data. The primary time base oscillator period, test time base oscillator period, quiescent current, low voltage detection level, sensor threshold level shall be measured and recorded as variables data for the first article samples and surveillance units (see 4.5.23).

3.8 Surveillance units. Quantities shall be as specified in the contract.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Output terminals. The output terminal well shall not contain any potting material flashing nor shall the terminal pins have any evidence of potting material adhering to the pins extending from the flat bottom section of the well. The terminal pins shall not have any evidence of exposed base metal.

3.11 Qualification testing of electrical components. Component level qualification requirements for integrated circuits, thick film resistor networks and discrete transistors, resistors, diodes and capacitors will be defined in 9317646.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply, and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample, as designated by the Contracting Officer, for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the parts/assemblies listed below in the quantities indicated:

Part Description	Drawing	Quantity
Electronic Assembly	9287150-1 9287150-2	50 See Table 1
Electronic Assembly	9287151-1 9287151-2	50 See Table 1
Printed Wiring Board Assembly (with AD switch)	9287138	10
Printed Wiring Board	9287176	10
Magnetometer Assembly (Prior to potting)	9287146	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection. See MIL-A-48078. The first article sample shall be rejected if any part, sub-assembly or assembly fails to comply with any of the examinations or tests specified in this specification or any of the requirements of the applicable drawings. In addition, the first article samples shall be rejected if the ballistic test sample fails as described below:

P/N 9287150-1 & -2

a. Any one (1) electronic assembly functions in any mode after expiration of the self-destruct time out (Hazardous dud).

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b. Five (5) or more electronic assemblies function prior to the minimum self-destruct time or prior to being subjected to a valid target signature (early).

c. A total of three (3) or more electronic assemblies fail to function when subjected to a valid target signature.

d. A total of four (4) or more electronic assemblies fail to function in any mode (Non-hazardous dud).

P/N 9287151-1 & -2

a. Any one (1) electronic assembly functions in any mode after expiration of the self-destruct time out (Hazardous dud).

b. Five (5) or more electronic assemblies function prior to the minimum self-destruct time or prior to being subjected to a valid target signature (early).

c. A total of three (3) or more electronic assemblies fail to function when subjected to a valid target signature.

d. A total of four (4) or more electronic assemblies fail to function in any mode (Non-hazardous dud).

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Electronic Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Electronic Assembly (Dwg. 9287150-1)				
	Examination for defects	50		3.2	4.4.2.27
	Ballastic test	Note 1		3.4.2	4.5.25
	(Dwg. 9287150-2)				
	Examination for defects	Note 1		3.2	4.4.2.27
	Ballistic test	Note 1		3.4.2	4.5.25
	Electronic Assembly (Dwg. 9287151-1)				
	Examination for defects	50		3.2	4.4.2.28
	Ballistic test	Note 2		3.4.2	4.5.25
	(Dwg. 9287151-2)				
	Examination for defects	Note 2		3.2	4.4.2.28
	Ballistic test	Note 2		3.4.2	4.5.25
	Printed Wiring Board Assy. (Dwg. 9287138				
	Examination for defects	10		3.2	4.4.2.25/4.4.2.26
	Magnetometer Assy. (Dwg. 9287146)				
	Examination for defects	10		3.2	4.4.2.3

NOTES: (1) Quantity required for five (5) rounds per Dwg. 9317597 requirements.
(2) Quantity required for five (5) rounds per Dwg. 9317598 requirements.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B (AR)

PARAGRAPH	TITLE: Electronic Assembly		SHEET 2 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Printed Wiring Board (Dwg. 9287176)				
	Examination for defects	10		3.2	4.4.2.22
	Solderability	10		3.2	4.4.2.22

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of Electronic Assemblies and Sub-assemblies shall contain:

a. Batteries and/or power supply assemblies bearing the same lot interfix number from one manufacturer.

b. Triple operational amplifiers bearing the same lot interfix number from one manufacturer.

c. Integrated circuit, CMOS, target sensors bearing the same lot interfix number from one manufacturer.

d. Integrated circuit, CMOS, timers bearing the same lot interfix number from one manufacturer.

e. Electronic assemblies of one part number only.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Cover, Front		9277745
			NEXT HIGHER ASSEMBLY 9287150/9287151

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Diameter datum A		0.40%	3.2	Gage
103	Height		0.40%	3.2	Gage
104	Spherical radius		0.40%	3.2	Comparator/Gage
105	True position of OD to datum A		0.40%	3.2	Gage
Minor					
201	Thickness		0.65%	3.2	Gage
202	Finish missing or incomplete		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Core, Toroid		9287094
			NEXT HIGHER ASSEMBLY 9292216

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Internal construction (notes 2D of dwg)	32	Note 1	3.2	Visual and Microscope
102	Weld strength (note 2E of dwg)	32	Note 1	3.2	Visual and manual
Minor	None defined				

NOTES:
(1) Reject lot if two (2) defective found.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Magnetometer Assembly (before potting)				9287146
					NEXT HIGHER ASSEMBLY 9287138
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Winding self-bonded and not unwinding		0.40%	3.2	Visual
102	Coating over entire winding and core		0.40%	3.2	Visual
103	Lead(s) improperly soldered to terminal(s)		0.40%	3.5.5	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Capacitor, Fixed, Electrolytic, Tantalum, Solid				9287096
					NEXT HIGHER ASSEMBLY 9287138
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current		0.40%	3.2	MIL-STD-202, Method 302
103	Weld strength (two sides)	8	Note 1	3.2	Pull Tester
104	Internal workmanship	20	Note 2	3.2	9292220
105	Breakdown voltage	10	Note 2	3.2	9292220
106	Dissipation factor		0.50%	3.2	MIL-STD-202, Method 306
107	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
108	Sleeve missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Reject for two (2) or more defective.
(2) Reject for three (3) or more defective.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Capacitor, Fixed, Electrolytic, Tantalum, Solid		9287126
			NEXT HIGHER ASSEMBLY 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current		0.40%	3.2	Mil-STD-202, method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Weld strength (two sides)	8	Note 1	3.2	Pull Tester
105	Internal workmanship	20	Note 2	3.2	9292220
106	Breakdown voltage	10	Note 2	3.2	9292220
107	Sleeve missing		0.40%	3.2	Visual
108	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Reject for two (2) or more defective.
(2) Reject for three (3) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Capacitor, Fixed Plastic Dielectric				9287104
					NEXT HIGHER ASSEMBLY
					9287138
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

(1) Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Capacitor, Fixed, Ceramic, Dielectric-Axial Lead				9287177
					NEXT HIGHER ASSEMBLY 9287138
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	(Leakage current) insulation resistance		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability, 2 leads	10	Note 1	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Reject for one (1) or more defectives.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Resistor		RCR07GXXXX/MIL-R-39008/
			NEXT HIGHER ASSEMBLY
			9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
102	Resistance		0.40%	3.2	MIL-R-39008/1
Minor					
201	Identification and marking		0.65%(a)	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(a) Refer to drawing for resistance value.
(1) Reject for one (1) or more defective.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Resistor, Fixed Film		9287099-XXXX
			NEXT HIGHER ASSEMBLY 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
Minor 201	Evidence of poor workmanship		0.65%	3.6	Visual
202	Identification and marking		0.65%	3.2	Visual

NOTES:

(1) Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Resistor, Fixed Film		9287099-1611
			NEXT HIGHER ASSEMBLY 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Resistance		0.40%	3.2	MIL-R-39008
102	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual
202	Identification and marking		0.65%	3.2	Visual

NOTES:

(1) Reject for one (1) or more defective.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH 4.4.2.11	TITLE Transistor (JAN 2N1907A)		SHEET 1 OF 1		DRAWING NUMBER 9287122-17 MIL-S-19500/291
					NEXT HIGHER ASSEMBLY 9287138
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined				
Major					
101	Wire material (See Note 2 of dwg.)	32	Note 1	3.2	Visual
102	Wire bond (See Note 2 of dwg.)	32	Note 1	3.2	Pull Tester (Note 2)
103	MIL-S-19500 Compliance			3.2	(Note 3)
104	Solderability (3 leads)	10	Note 1	3.5.5.1	(Note 4)
105	Leakage Current (ICE)			3.2	Certify to dwg 9287122
106	Internal Visual			3.2	Certify to dwg 9287122
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Reject for one (1) or more defective.
(2) Pull all wires on each device.
(3) Visual evidence of JAN marking.
(4) Use applicable inspection method per para. 3.5.5.1.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Transistor (JAN2N2222A)	1 of 1	9287122-2/ MIL-S-19500/255
			NEXT HIGHER ASSEMBLY: 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire material (See Note 2 of dwg)	32	Note 1	3.2	Visual
102	Wire bond (See Note 2 of dwg)	32	Note 1	3.2	Pull Tester (Note 2)
103	MIL-S-19500 Compliance			3.2	(Note 3)
104	Solderability (3 leads)	10	Note 1	3.5.5.1	(Note 4)
105	Leakage Current (ICE)			3.2	Certify to dwg 9287122
106	Internal Visual			3.2	Certify to dwg 9287122
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Reject for one (1) or more defective.
(2) Pull all wires on each device.
(3) Visual evidence of JAN marking.
(4) Use applicable inspection method per para. 3.5.5.1.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.13	Network, Resistor-Fixed and Adjustable		9292219
			NEXT HIGHER ASSEMBLY 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Resistance (all)		0.40%	3.2	MIL-STD-202, Method 303
102	Resistance-temperature characteristic (Note 2)	32	Note 1	3.2	MIL-STD-202, Method 304
103	Solderability all leads	10	Note 1	3.2	MIL-STD-202, Method 208
104	Evidence of cracks, chips, lead fatigue or other physical damage		0.40%	3.2	Visual
105	Approved part/vendor verification	5	Note 1	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Reject for one (1) or more defective.
(2) Required for vendor qualification only.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.14	Network, Resistor-Capacitor		9292137
			NEXT HIGHER ASSEMBLY 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance (all)		0.40%	3.2	MIL-STD-202, Method 305
102	Resistance (all)		0.40%	3.2	MIL-STD-202, Method 303
103	Resistance-temperature characteristic (Note 2)	32	Note 1	3.2	MIL-STD-202, Method 304
104	Solderability (all leads)	10	Note 1	3.2	MIL-STD-202, Method 208
105	Evidence of cracks, chips, lead fatigue or other physical damage		0.40%	3.2	Visual
106	Approved part/vendor verification	5	Note 1	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Reject for one (1) or more defective.
(2) Require for vendor qualification only.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.15	Network, Resistor-Capacitor		9292175
			NEXT HIGHER ASSEMBLY 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance (all)		0.40%	3.2	MIL-STD-202, Method 305
102	Resistance (all)		0.40%	3.2	MIL-STD-202, Method 303
103	Resistance-temperature characteristic (Note 2)	32	Note 1	3.2	MIL-STD-202, Method 304
104	Solderability (all leads)	10	Note 1	3.2	MIL-STD-202, Method 208
105	Evidence of cracks, chips, lead fatigue or other physical damage		0.40%	3.2	Visual
106	Approved part/vendor verification	5	Note 1	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Reject for one (1) or more defective.
(2) Require for vendor qualification only.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298512
4.4.2.16	A.D. Switch Assembly				NEXT HIGHER ASSEMBLY 9287138
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Switch function seven (7) contact closures minimum each direction		Note 1	3.2	4.5.17.1, Note 2
102	Contact closure duration 0.60, seconds max.		Note 1	3.2	4.5.17.1, Note 2
103	Insulation resistance		Note 1	3.2	4.5.17.2, Note 2
104	Assembly damaged, distorted or incomplete		0.40%	3.2	Visual
105	Ribbon improperly welded to can		0.40%	3.2	Visual
106	Epoxy seal missing or incomplete		0.40%	3.2	Visual
107	Solderability, 2 leads (Note 3)			3.5.5.1	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.6	Visual
NOTE 1:	All inspection lots of AD switches shall be subjected to 100% production screening for proper function (Major 101, 102, and 103) with all defectives removed.				
NOTE 2	A defect is defined as: a switch with less than 7 momentary contact closure in each direction while being rotated 360° clockwise and counter clockwise contact closure duration exceeding 0.60 second maximum; a switch with an insulation resistance between switch terminals, less than two (2) megohms. If any switch has any closure that exceeds 1.20 seconds, the lot shall be rejected. The lot shall also be rejected if any switch has more than two contact closures between 0.60 and 1.20 seconds.				
NOTES: NOTE 3	Verify double dip operation each shift.				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.17	Semiconductor Device, Diode, Type JAN IN645-1		9287168
			NEXT HIGHER ASSEMBLY 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Reverse leakage current	1/	0.40%	3.2	MIL-STD-750, Method 4016dc
102	Forward voltage	10	0.40%	3.2	MIL-STD-750, Method 4011
103	Solderability	10	0-1	3.2	MIL-STD-750, Method 2026
104	Evidence of JAN marking	10	0-1	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

1/ Color band end shall be used as cathod end during reverse current test, reverse current test is used to confirm that the color band is on cathod end or diode.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.18	Triple Operational Amplifier	1	1	MIL-A-63242
				NEXT HIGHER ASSEMBLY 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Contractor shall furnish evidence of having satisfactorily met inspection requirements of MIL-A-63242			3.2	Certification
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287173
4.4.2.19	Integrated Circuit, CMOS, Target Sensor				NEXT HIGHER ASSEMBLY 9287138
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Contractor shall furnish evidence of having satisfactorily met inspection requirements of drawing 9287173.			3.2	Certification
Minor 201	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-6324B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287165
4.4.2.20	Integrated Circuit, CMOS, Timing				NEXT HIGHER ASSEMBLY 9287138
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Contractor shall furnish evidence of having satisfactorily met inspection requirements of drawing 9287165.			3.2	Certification
Minor 201	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH 4.4.2.21	TITLE Printed Wiring Board		SHEET 1 OF 1		DRAWING NUMBER 9287176
					NEXT HIGHER ASSEMBLY 9287138
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hole diameters	5	.40%	3.2	Gage (Note 2)
102	Solderability	5	.40%	3.5.5.1	4.5.22
103	Plated thru hole integrity	5	.40%	3.2	Visual
104	Compliance of board to dwg. 9287176		Lot		Certification to Note 1A of drawing
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) The AQL applies to the number of holes per board. If any board exceeds this criteria the lot shall be rejected.
(2) Hole diameters undersized by .002 inch are acceptable if caused by reflow build up.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63421B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.22	Terminal		9287185
			NEXT HIGHER ASSEMBLY 9287138

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Solderability	20	Note 1	3.2	MIL-STD-202, Method 208
Minor 201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: (1) Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Printed Wiring Board Assembly (Configured for 9287150)				9287138
					NEXT HIGHER ASSEMBLY 9287150-1 & -2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship		Note 1	3.5.5	Visual/4.5.22
102	Conformal coating missing from noted components		0.40%	3.2	Visual
103	Any component damaged, missing or improperly assembled		0.40%	3.2	Visual
104	Battery lead welds(s) broken		Note 2	3.2	Visual (5X)
Minor					
201	Height from board to top of battery mold		0.65%	3.2	Gage
202	Height of leads/solder on bottom of board		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Accept/reject per Appendix para. 50.
(2) Lot size is 500 units (maximum). AQL is 315-0-1.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	Printed Wiring Board Assembly (Configured for 9287151)				9287138
					NEXT HIGHER ASSEMBLY 9287151-1 & -2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship		Note 1	3.5.5	Visual/4.5.22
102	Conformal coating missing from noted components		0.40%	3.2	Visual
103	Any component damaged, missing or improperly assembled		0.40%	3.2	Visual
104	Battery lead welds(s) broken		Note 2	3.2	Visual (5X)
Minor					
201	Height from board to top of battery mold		0.65%	3.2	Gage
202	Height of leads/solder on bottom of board		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Accept/reject per Appendix para. 50.
(2) Lot size is 500 units (maximum). AQL is 315-0-1.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.25	Electronic Assembly				9287150-1 & 2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	(Note 1)				
1	SD Timeout high		100%	3.5.2.4.1	4.5.7 4.5.26
2	SD Timeout high - real time test Hot temperature (Note 4)		4-0-1	3.5.2.4.1	4.5.7
	Cold temperature (Note 5)		4-0-1	3.5.2.4.1	4.5.7
	Ambient temperature	16	Note 3	3.5.2.4.1	4.5.7
3	Max PTB oscillator peroid		100%	3.5.2.1	4.5.2
4	Correlation between SD jumper position and part number		100%	3.2	Visual
Special	(Note 2)				
a	Max TTB oscillator peroid		100%	3.5.2.1	4.5.3
b	Oscillator check circuit function		100%	3.5.2.5	4.5.9
c	Low voltage circuit trigger level detect function		100%	3.5.2.3	4.5.6
d	Self-sterilization circuit function		100%	3.5.2.8	4.5.12
NOTE (1)	Report all critical defects per 4.5.21.				
NOTE (2)	See para 6.7.				
NOTE (3)	16-0-1 Critical 16-0-2 Dud 32-1-2 Dud (Double Sample) 16-1-3 Early 32-2-3 Early (Double Sample)				
NOTES:					

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25 Cont'd.	Electronic Assembly	2 OF 2	9287150-1 & 2
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Min PTB oscillator peroid	4.4.3	1	3.5.2.1	4.5.2
102	Min TTB oscillator peroid	4.4.3	1	3.5.2.1	4.5.3
103	Arming delay/sensor enable	4.4.3	1	3.5.2.2	4.5.4/4.5.5
104	SD timeout low	4.4.3	1	3.5.2. 4.1	4.5.7
105	AD switch mode (9287150-2, Ambient only)	Note 6		3.5.2.6	4.5.10
106	Shorting bar continuity	4.4.3	1	3.5.6	4.5.16
107	Output energy	4.4.3	1	3.5.2.7	4.5.11
108	Target sensor function	4.4.3	1	3.5.3	4.5.13
109	Quiescent current	4.4.3	1	3.5.1	4.5.1
110	Potting material characteristics	Note7		3.3.2	4.5.15
111	Excessive voids/bubbles in potting		0.40%	3.3.2	Visual/Gage
112	Spherical radius profile		0.40%	3.2	Gage
113	Depth of "D" hole		0.40%	3.2	Gage
114	Diameter of "D" hole		0.40%	3.2	Gage
115	True position of "D" hole to center hole		0.40%	3.2	Gage
NOTE (6)	Lot Size / Sample: 0-500 / 12-1-2; 501-3200 / 45-2-3; 3201-10,000 / 68-3-4; 10,001-35,000 / 105-4-5				
	NOTE (7): Sample in accordance with 4.4.3.2.		Potting	process	shall be verified daily

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.25 Cont'd.	Electronic Assembly				9287150-1 & 2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
116	True position of OD to center hole		0.40%	3.2	Gage
117	Height from flange to top of "D" hole		0.40%	3.2	Gage
118	Depth from the bottom of "D" hole to the lower shoulder of the mid-section center hole		0.40%	3.2	Gage
119	ID (min) of the mid section center hole		0.40%	3.2	Gage
120	ID (min) of the top section center hole		0.40%	3.2	Gage
121	Terminal(s) extending above spherical surface		0.40%	3.2	Visual
122	Presence of AD switch (9287150-2 only)		0.40%	3.2	Visual
123	Read Time S^ timeout low - Dud and early				4.5.26
	Hot temperature (Note 4)	8-1-3 16-2-3		3.5.2. 4.1	4.5.26
	Cold temperature (Note 5)	8-1-3 16-2-3		3.5.2. 4.1	4.5.26
	Ambient (Note 3)	16		3.5.2. 4.1	4.5.26
124	Ballistic test	27	Note 8	3.4.2	4.5.25
125	One or both battery cells activated		100%	3.5.4	4.5.14
Minor					
201	Width of test pad hole		0.65%	3.2	Gage
202	Length of test pad hole		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: (4) Same unit shall be used for these hot temperature tests.
(5) Same units shall be used for these cold temperature tests.
(8) Test for information only.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.26	Electronic Assembly				9287151-1 & 2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	(Note 1)				
1	SD timeout high		100%	3.5.2. 4.1	4.5.7
2	SD timeout high - real time test				4.5.26
	Hot temperature (Note 4)		4-0-1	3.5.2. 4.1	4.5.7
	Cold temperature (Note 5)		4-0-1	3.5.2. 4.1	4.5.7
	Ambient temperature	16	Note 3	3.5.2. 4.1	4.5.7
3	Max PTB oscillator peroid		100%	3.5.2.1	4.5.2
4	Correlation between SD jumper position and part number		100%	3.2	Visual
Special	(Note 2)				
a	Max TTB oscillator peroid		100%	3.5.2.1	4.5.3
b	Oscillator check circuit function		100%	3.5.2.5	4.5.9
c	Low voltage circuit trigger level detect function		100%	3.5.2.3	4.5.6
d	Self-sterilization circuit function		100%	3.5.2.8	4.5.12
NOTE (1)	Report all critical defects per 4.5.21.				
NOTE (2)	See para 6.7.				
NOTE (3)	16-0-1 Critical				
	16-0-2 Dud				
	32-1-2 Dud (Double Sample)				
	16-1-3 Early				
NOTES:	32-2-3 Early (Double Sample)				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.26 Cont'd.	Electronic Assembly				9287151-1 & 2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Min PTB oscillator peroid	4.4.3.1		3.5.2.1	4.5.2
102	Min TTB oscillator peroid	4.4.3.1		3.5.2.1	4.5.3
103	Arming delay/sensor enable	4.4.3.1		3.5.2.2	4.5.4/4.5.5
104	SD timeout low	4.4.3.1		3.5.2.4.1	4.5.7
105	AD switch mode (9287150-2, Ambient only)	Note 6		3.5.2.6	4.5.10
106	Shorting bar continuity	4.4.3.1		3.5.6	4.5.16
107	Output energy	4.4.3.1		3.5.2.7	4.5.11
108	Target sensor function	4.4.3.1		3.5.3	4.5.13
109	Quiescent current	4.4.3.1		3.5.1	4.5.1
110	Potting material characteristics	Note 7		3.3.2	4.5.15
111	Excessive voids/bubbles in potting		0.40%	3.3.2	Visual/Gage
112	Spherical radius profile		0.40%	3.2	Gage
113	Depth of "D" hole		0.40%	3.2	Gage
114	Diameter of "D" hole		0.40%	3.2	Gage
115	True position of "D" hole to center hole		0.40%	3.2	Gage

NOTE (6)

Lot Size	Sample
0-500	12-1-2
501-3200	45-2-3
3201-10,000	68-3-4
10,001-35,000	105-4-5

NOTE (7) Sample in accordance with 4.4.3.2. Potting process shall be verified daily.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63241B(AR)

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.26 Cont'd.	Electronic Assembly		9287151-1 & 2
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
116	True position of OD to center hole		0.40%	3.2	Gage
117	Height from flange to top of "D" hole		0.40%	3.2	Gage
118	Depth from the bottom of "D" hole to the lower shoulder of the mid-section center hole		0.40%	3.2	Gage
119	ID (min) of the mid section center hole		0.40%	3.2	Gage
120	ID (min) of the top section center hole		0.40%	3.2	Gage
121	Terminal(s) extending above spherical surface		0.40%	3.2	Visual
122	Presence of AD switch (9287151-2 only)		0.40%	3.2	Visual 4.5.26
123	Real time SD timeout low - Dud and early				
	Hot temperature (Note 4)	8-1-3 16-2-3		3.5.2. 4.1	4.5.26
	Cold temperature (Note 5)	8-1-3 16-2-3		3.5.2. 4.1	4.5.26
	Ambient (Note 3)	16		3.5.2. 4.1	4.5.26
124	Ballistic test	27	Note 8	3.4.2	4.5.25
125	One or both battery cells activated		100%	3.5.4	4.5.14
Minor					
201	Width of test pad hole		0.65%	3.2	Gage
202	Length of test pad hole		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: (4) Same unit shall be used for these hot temperature tests.
(5) Same units shall be used for these cold temperature tests.
(8) Test for information only.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables and below.

4.4.3.1 Ambient (75°F + 10°F), Hot (145°F + 5°F) and Cold (-25°F -5°F) temperature functioning tests. The table below shall be used for the selection of the lot acceptance sample sizes for ambient, hot and cold temperature testing of lots of electronic assemblies.

INSPECTION LOT SIZE	SAMPLE SIZE	* AMB	HOT	COLD	ACCEPT - REJECT CRITICAL/SPECIAL DEFECTS	MAJOR DEFECTS
0-150	20	8	6	6	0-1	0-1
151-500	32	12	10	10	0-1	0-1
501-3200	125	45	40	40	0-1	1-2
3201-10,000	200	68	66	66	0-1	2-3
10,000-35,000	315	105	105	105	0-1	3-4

The units shall be tested for all the functional parameters (under critical, special, major) listed in 4.4.2.27, 4.4.2.28. Reject number is the number of defective units that would reject the lot. Since all the units have to be 100% inspected at ambient for critical and special defects, the quantities list under * do not have to be reinspected for critical and special defects at ambient temperature (see 4.5.18).

4.4.3.2 Potting material testing. The potting material shall be checked for tensile strength, and hardness in accordance with drawing 9287148. Two test samples shall be molded per the drawing for each batch of mixed material or each 1,000 electronic assemblies assembled during continuous production, or each days production, whichever comes first. One test sample will be used for testing and one retained per the drawing. Identification shall be maintained between the sample and assemblies. Failure of the test specimen to comply with requirements shall result in rejection of the lot of electronic assemblies assembled from the batch of potting material (see 4.5.15).

4.4.3.3 Inspection criteria. After three (3) consecutive lots have successfully passed 100% inspection, the contractor may switch to sample inspection at an AQL .25% in accordance with MIL-STD-105D. The above one hundred percent (100%) inspection shall be reinstated if test sample fails to comply with the above acceptance criterial level.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is identified in the 'Paragraph Reference/Inspection Method' column in Classification of Defects and Tests forms beginning with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and section 6.3 herein.

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4.4.5 Dimensional control of molded parts. In place of normal sampling and inspections associated with the Classification of Defects and after a curing time for the parts has been approved (see note 1), the contractor may, with the approval of the Government, select a sample of three (3) parts (as molded) from each cavity to be fully inspected dimensionally toqualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in pertinent paragraphs 4.4.2.1 through 4.4.2.24 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the contrl of the cavity, the cavity producing the defective partshall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAT-M for approval prior to manufacture of equipment. The note subparagraphs identify this paragraph.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAT-M. When a new cavity is substituted, an old cavity reworked, or the material or cycle time hanged, a new curing time shall be established and approved (see 6.4).

NOTE 2: Those features of a part which are subsequently machined or otherwise worked upon to meet its drawing requirements shall be inspected in accordance with the classification of defects.

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4.5 Test methods and procedures. Only Government approved test equipment and test fixtures shall be connected to electronics assemblies (electronic assembly) under test to verify that it complies with the requirements specified herein. Increasing the PTB oscillator frequency to reduce test time is acceptable, provided the higher frequency oscillator signal maintains an approximate 50% duty cycle and has a peak amplitude frequency oscillator signal which does not cause or mask malfunctions in the functional operation of the electronic assembly; the higher frequency oscillator signal is well within the capability of the test equipment to monitor and control it with an accuracy of one cycle; the actual procedure used to increase the frequency of the signal is approved by the Government before it is implemented. When the PTB oscillator frequency is increased, the TTB oscillator frequency is to be proportionately increased or the TTB oscillator inhibited as applicable for the test being implemented. All testing shall be performed, unless otherwise specified, at a supply voltage of 6.75 ± 0.15 volts, an ambient temperature of 75°F ± 10°F, and a relative humidity of 60 ± 20%. Also, the rise time of the supply voltage applied to the electronic assembly under test shall not be less than 5 milliseconds nor greater than 200 milliseconds.

4.5.1 Quiescent current test. With the electronic assembly under test connected to the test fixture and with the detonator terminals connected to ground through a 3.9 ± 5% ohm resistor, measure the quiescent current of the electronic assembly. The measured current shall be observed for compliance with the requirement specified in 3.5.1. This is a non-destructive test.

4.5.2 Primary time base (PTB) period test. With the electronic assembly under test connected to the test fixture and with both the primary and test time base circuits oscillating at their natural frequency, measure the period of the PTB oscillator at test point T.P.4. Observe the measured period for compliance with the requirements specified in 3.5.2.1 for the corresponding temperatures and voltages under which this test is conducted. This is a non-destructive test.

4.5.3 Test time base (TTB) period test. With the electronic assembly under test connected to the test fixture and with both the primary and test time base oscillators operating at their natural frequency, measure the period of the TTB oscillator at test point T.P.5. Observe the measured period for compliance with the requirements specified in 3.5.2.1 for the corresponding temperatures and voltages under which this test is conducted. This is a non-destructive test.

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4.5.4 Arming delay signal test. With the electronic assembly under test connected to the test fixture, verify that the arming delay signal occurs at the PTB clock count and temperature specified in 3.5.2.2. This is a non-destructive test.

4.5.5 Sensor enable test. With the electronic assembly under test connected to the test fixture, verify that sensor enable occurs at the PTB clock count, temperature and voltage specified in 3.5.2.2. Make the verification by incorporating, as a minimum, the following checks in the test procedures.

a. At the PTB clock count of 3 (± 2) less than that required for sensor enable, activate magnetic sensor circuit and confirm that the detonator firing circuit does not turn on.

b. At the PTB clock count of 3 (± 2) greater than that required for sensor enable, activate magnetic sensor circuit and confirm that the detonator firing circuit is turned on. This is a non-destructive test.

4.5.6 Low voltage detect test. With the electronic assembly under test connected to the test fixture, verify that the low voltage detect circuit initiates the turn on of the detonator firing circuit at the temperature and voltage specified in 3.5.2.3. For this test it is required that both the PTB and TTB oscillators be operating at their natural frequency, and that the initial supply voltage be 6.75 ± .15 volts and linearly reduced at a rate of 6.7 (max.) volts/second until the low voltage detect circuit is triggered. This is a non-destructive test.

4.5.7 Self-destruct signal test (Dwg. 9287150). With the electronic assembly under test connected to the test fixture, verify that the self-destruct signal, that initiates the turn on of the detonator firing circuit, occurs at the PTB count specified in 3.5.2.4.1. This is a non-destructive test.

4.5.8 Self-destruct signal test (Dwg. 9287151). With the electronic assembly under test connected to the test fixture, verify that the self-destruct signal, that initiates the turn on of the detonator firing circuit, occurs at the PTB count specified in 3.5.2.4.2. This is a non-destructive test.

4.5.9 PTB oscillator circuit check. With the electronic assembly under test connected to the test fixture and both the PTB and TTB oscillator circuits operating at their natural frequencies, verify that the oscillator check circuit is operational by confirming that the oscillator check circuit initiates the turn on of the detonator firing circuit within 2 TTB periods after the PTB oscillator is stopped. Observation shall be made for compliance with requirements of 3.5.2.5. This is a non-destructive test.

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4.5.10 Anti-disturbance function mode. After the output of 176 pulses, with the electronic assembly under test connected to the test fixture so that its plane makes an approximate 45 degree angle with the horizontal, the electronic lens assembly shall be moved thru an arc of 90 degrees maximum about the cylindrical axis. Observation shall be made that the detonator firing circuit turns on before the 90 degree rotation is complete. Observation shall be made for compliance with the requirements of 3.5.2.6. This is a non-destructive test. All defects shall be rerun at the end of the lot acceptance test, once, to verify the defect. If a unit functions properly upon retest, it shall be classified as an acceptable unit."

4.5.11 Firing circuit energy. With the electronic assembly under test connected to the test fixture, verify that the parallel detonators firing circuits Pin E2 meet the requirements specified in 3.5.2.7. This is a non-destructive test.

4.5.12 Self-sterilization function. With the electronic assembly under test connected to the test fixture and the detonator firing circuitan open circuit, verify that the voltage of the firing capacitor meets the requirements of 3.5.2.8. This is a non-destructive test.

4.5.13 Magnetic target sensor functioning test. The potted electronic assembly shall be connected to approved test equipment and tested for proper functioning of the magnetic sensor circuitry. The electronic assembly shall be tested at the magnetic latitude extremes specified on drawing 9287153 for the ambient all-fire and no-fire levels and firing angle specified on the drawing. High leel cut-off functioning and spurious outputs shall also be checked at this assembly level. Observation shall be made for compliance with the applicable requirements of 3.5.3. Non-destructive test.

NOTE: "When testing for the magentic function requirements specified in Table 1 of Dwg. 9287153, the acceptance test signal amplitude for values specified as" "shall be between the value specified plus two (2) millioersteds, the acceptance test limit for values specified as " "shall be between the value specified minus two (2) millioersteds."

4.5.14 Non-activated voltage. Each battery cell shall be checked for conformance with the requirements of 3.5.4 when measured by a 10 megohm input impedance volt meter. Measurement shall be made of each cell individually in the potted electronic assembly. Non-destructive test.

4.5.15 Potting material testing. Tests of the potting material shall be done in accordance with drawing 9287148 and 4.4.3.2 herein. Observation shall be made for compliance with requirements of 3.3.2 and 4.4.3.2.

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4.5.16 Shorting bar resistance. The continuity - resistance of each shorting bar circuit shall be measured, using an approved ohmmeter, between the test points E3 and TP9. Observation shall be made for compliance with requirement of 3.5.6. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.17 Switch assembly (Anti-disturbance switch).

4.5.17.1 AD switch function and closure time duration. The switch assembly shall be placed in an approved fixture. The fixture shall be capable of rotating the switch assembly through 360° on its cylindrical axis, while this axis is held at 15 ± 2° referenced to the horizontal, for both CW and CCW directions of rotation. Verify that the functional requirements of the drawing are met. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.17.2 Insulation resistance of AD switch. The switch assembly when in a suitable fixture shall be measured for insulation resistance in the reset position using an approved megohmmeter. Verify that the switch assembly does meet the requirements of the drawing. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.17.3 Plating of AD switch can. Five (5) cans shall be randomly selected from each plating batch. The specimens shall be cross-sectioned through the axial center of the can, polished and examined microscopically to measure the plating thickness. If the total average plating thickness for the five cans fails to meet the requirements as specified on the drawing, the sample shall be classed defective and the lot shall be rejected. (Destructive test).

4.5.18 Ambient, hot and old temperature functional testing. The sample of electronic assemblies specified in para. 4.4.3.1 shall be subjected to all functional tests at the operating temperature extremes. One-third of the sample specified shall be functionally tested atambient (75°F ± 10°F) and one-third at-25°F (+0°, -3°) and one-third at 145°F (+3°, -0°), (see 4.4.3.1). The units shall be conditioned at the above temperatures for two (2) hours prior to the test. Testing shall be performed with the unit in the temperature chamber. Approved functional test equipment shall be used. Acceptable threshold levels are specified on drawing 9287153, Table 1 for the temperature extremes. Acceptance limits for the digital circuitry (SD timing control) at the temperature extremes are specified on drawing 9287153, Table II.

4.5.19 Printed wiring board cleaning procedures. Printed wiring board assembly cleaning procedures shall be documented and supplied to the Government (see 6.6). Verification of adherence to the cleansing procedure shall be done on a daily basis (Ref. Data Item DI-T-5372 (Tailored).

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4.5.20 Integrated circuit handling procedures. CMOS integrated circuits require care in handling to avoid damage due to static discharge. The contractor shall document his handling procedure for the CMOS integrated circuits and supply the procedure to the Government (see 6.6). Adherence to the handling procedure shall be verified daily. (Ref. DI-R-5372 (Tailored)).

4.5.21 Data requirements. The contractor shall submit a quality inspection test, demonstration and evaluation report in accordance with Data Item Description DI-R-1724. It shall include all detailed inspection and test results. Test data reports shall be submitted to the technical agency listed in 6.4.

4.5.22 Quality assurance soldering plan. The QA soldering plan as noted in 3.5.5.2 shall be submitted for Government approval (see 6.6). (Ref. DI-T-5372 (Tailored)).

4.5.23 Variables data. The contractor shall submit variables data in accordance with Data Item Description DI-R-1724 for the First Article samples and surveillance units. It shall include all detailed inspection and test results with respect to the following parameters: Primary Time Base Oscillator Period, Test Time Base Oscillator Period, Quiescent Current, Low Voltage Detection Level, Magnetic Sensor Threshold Level. Test data reports shall be submitted to the technical agency listed in 6.4.

4.5.24 Surveillance units. The electronic and battery assemblies designated for surveillance are to be stored at the contractors facility until contract completion and disposition is made. The units are to be identified and serialized in a manner to provide traceability to month and year of manufacture.

4.5.25 Ballistic test. Electronic assemblies shall be capable of performance as stated herein after being subjected to the ballistic environment of the 155MM M109A1 Howitzer, zone 8 and conditioned at 145°F. The electronic assemblies shall be forwarded to the load plant specified in the contract for assembly into non-hazardous test projectiles. Electronic assemblies, P/N 9287150-1 & -2, shall be loaded into Projectile, 155MM test, Non-hazardous, M718 type, Dwg. 9317597. Electronic assemblies, P/N 9287151-1 & -2, shall be loaded into Projectile, 155MM test, Non-hazardous, M741 type, Dwg. 9317598. Testing of the projectile and mines shall be in accordance with MIL-P-63238, para. 4.5.8 as applicable for the Non-hazardous type mines. Of the forty-five (45) mines of each type to be tested, nine (9) shall be randomly chosen and allowed to self-destruct per MIL-P-63238, para. 4.5.8.3.b.(1). The remaining thirty-six (36) shall be tested for sensor performance per MIL-P-63238, para. 4.5.8.3.b.(2).

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4.5.26 Self-Destruct (SD) mode destructive tests (Real Time). A randomly selected sample of electronics assemblies P/N 9287150-1, -2/9287151-1, -2, with simulated resistive load for the parallel detonators, shall have their batteries activated by appropriate test fixture and allowed to function in "Real Time" as follows:

a. Functioning shall be at hot, cold, and ambient.

b. SD functioning shall be part of lot acceptance tests. Observation shall be made for compliance with requirement of Dwg. 9287153. Units under test shall be conditioned at +145°F, -25°F, and +75°F.

c. Through use of fixtures and/or inclosures, the sample units shall be protected from radio frequency and magnetic field effect during the time interval covering activation of batteries through functioning.

d. If an electronic assembly fails to function by the maximum set self-destruct time, a valid target signature shall be applied (using a simulated prime target sensor signature coil). Any assembly failing to function (Non-hazardous dud) or functioning by sensor after SD time (Hazardous dud) shall be set aside for dismantling and failure analysis.

e. Event time shall be monitored and recorded.

5. PACKAGING.

5.1 Preservation and packaging. The electronic assemblies shall be preserved and packaged in accordance with MIL-STD-1169, Level C.

5.2 Packing. Packing of the assemblies shall be in accordance with MIL-STD-1169, Level C.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use in the M718/M741 System.

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6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

d. Certification of conformance.

e. Provisions for surveillance units.

6.3 Submission of inspection equipment designs for appoval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAT-M, Dover, New Jersey, 07801.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations of tests to Commander, ARRADCOM, ATTN: DRDAR-QAT-M, Dover, New Jersey, 07801.

6.5 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, ARRADCOM, ATTN: DRDAR-QAT-M, Dover, New Jersey 07801.

6.6 Submission of procedures. Submission of procedures specified in 4.4.3.2, 4.5.11, 4.5.20 and 4.5.22 shall be to Commander, ARRADCOM, ATTN: DRDAR-QAT-M, Dover, New Jersey 07801.

6.7 Special defect. A defect, other than critical that judgement and experience indicate may, depending upon tne degree of variance from the design requirement.

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or

b. Prevent performance of the tactical function of a major end item.

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SUPPLEMENTAL SOLDERING PROVISIONS

10 SCOPE: This appendix details supplementary soldering provisions to MIL-S-46844 for machine soldering and to MIL-S-45743 for hand soldering.

20 APPLICABLE DOCUMENTS: MIL-S-46844 and MIL-S-45743 listed in section 2 of this specification.

30 EXCEPTIONS AND CLARIFICATIONS:

30.1 Exceptions/clarifications to MIL-S-46844. The following exceptions/clarifications to reference MIL-S-46844 paragraphs shall apply.

Paragraph 3.6.2 flux application. Change to read "...even coated surface. Any thinner used shall be a product from the flux manufacturer recommended for the specified flux being utilized or a substitute approved by the technical agency."

Paragraph 3.6.5 The solder contact. Add: "Bonded components or clinched leads are acceptable restraints provided they are implemented in a manner to assure that the leads do not move during the solder cycle."

Paragraph 3.7.1 Cleaning. Change to read "...scraping shall not be used. Residual flux, activators and soils shall be removed to the degree approved by the technical agency using solvents or combination of solvents which will remove contaminants."

Paragraph 3.8 Resistivity of solvent extract. Delete

Paragraph 4.2.1 One hundred percent inspection. Delete and substitute the following: "4.2.1 Level of inspection. Inspection of soldered connections and associated wiring and leads shall be performed to the level specified in the classification of defects in the product specification. Table II of MIL-S-45743 provides a list of defects with reference to the applicable requirements."

Paragraph 4.2.2 Sampling inspection. Delete

Paragraph 4.3.1 Visual inspection. Change to read, "....A clear incandescent or fluorescent light should be used for detection of uncovered copper."

Paragraph 4.3.2 Cleaniness test. Delete

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30.2 Exceptions/clarifications to MIL-S-45743. The following exceptions/clarifications to reference MIL-S-45743 paragraphs shall apply.

Paragraph 3.3.1.2 Control of airborne contaminants. Delete the sentence, "Positive pressure shall .....between them."

Paragraph 3.3.2.4.5 Soldering irons and resistance soldering irons. Delete the sentence, "The soldering iron shall ....or standby condition.", and substitute the sentence, "The soldering iron shall be a soldering iron approved by the technical agency or a soldering iron that attains a temperature no more than 10°F (5.6°C) above the control temperature when in an idling or standby condition."

Paragraph 3.4.5.7 Printed board termination, Subparagraph 'b'. Change to read, "....is permissible. Leads shall not extend, beyond the terminal pad or printed wiring pattern, far enough to cause less than .015 inches of clearance between conductors."

Paragraph 3.4.5.7 Printed board termination, Subparagraph 'd'. Change to read, "When leads cannot be clinched per above, they shall extend through the printed board, be visible and properly soldered."

Paragraph 3.6.1 Cleaning the soldered connection. Change to read, "When leads cannot be clinched per above, they shall extend through the printed board, be visible and properly soldered."

Paragraph 3.4.5.8 Ribbon lead geometry. Add new sentence, "Ribbon lead termination from battery and lead assembly, shall comply with paragraph 3.4.5.7b."

Paragraph 3.5.1 Cleaning the soldered connection. Change to read, "...(within one (1) hour) using solvents or combination of solvents which will remove residual flux, activators, and other contaminants."

Paragraph 3.6.3 Resistivity of solvent extract. Delete

Paragraph 4.2.1 One hundred percent inspection. Delete and substitute the following: "4.2.1 Level of inspection. Inspection of soldered connections and associated wiring and leads shall be performed to the level specified in the classification of defects in the product specification. Table II of MIL-S-45743 provides a list of defects with reference to the applicable requirements."

Paragraph 4.2.2 Sampling inspection. Delete

Paragraph 4.3.1 Visual inspection. Change to read, "....A clear incandescent or fluorescent light should be used for detection of uncovered copper."

Paragraph 4.3.2 Cleanliness test. Delete

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40 SUPPLEMENTAL SOLDER WORKMANSHIP STANDARDS

40.1 General. The following solder workmanship standards supplement the workmanship requirements contained in MIL-S-46844C and MIL-S-45743E and are provided as additional aids in establishing the minimum acceptance standards for soldering. All examinations will be performed using a minimum 5 diopter lens. Suspected connections may be examined using 10x magnification to determine acceptability.

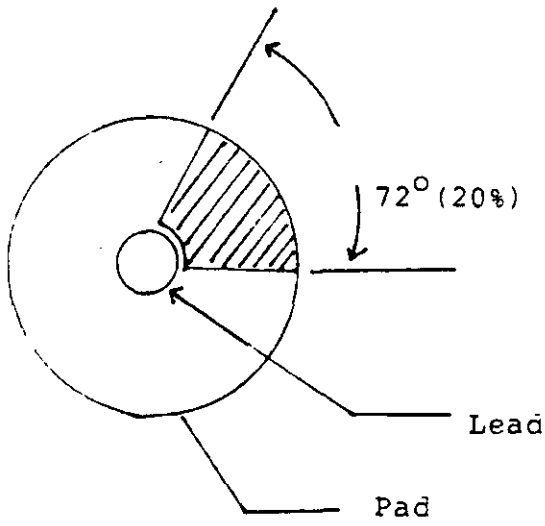
40.2 Definitions.

40.2.1 Voids. All holes, pits, and voids regardless of size are classified as voids.

40.2.1.2 Hole with solder plug. The total accumulative void area on each side of hole is less than 20% of the hole area and does not expose base metal. Also there are no voids through the board. Evidence of good wetting at the pad surface.

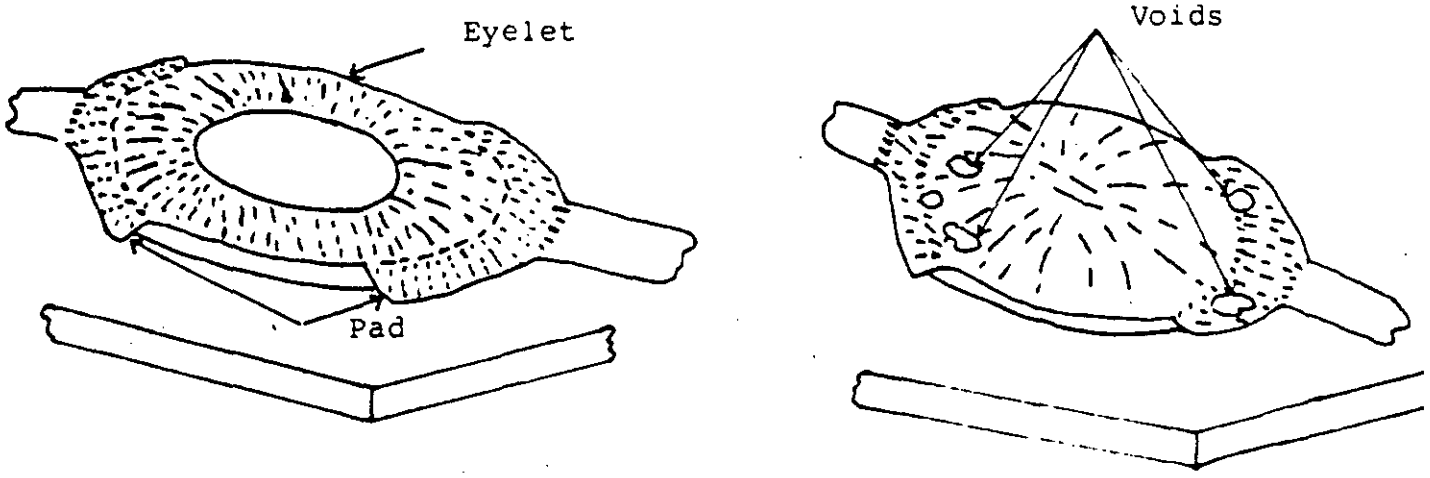
40.2.1.3 Hole with leadwire. For solder side and component side: the total accumulative void area is less than 20% of the lead circumference; does not visibly expose base metal; leads adjacent to voids exhibit good wetting and 100% solder coverage of the lead surface. There are no voids through the board. The lead and pad are wetted.

40.2.1.4 Twenty (20) % void. (For solder joint with a lead.) Toal void area equivalent to an area subtended by an arc equal to 20% of the lead circumference.



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40.2.1.5 Eyelet connection voids. The total cumulative area of all voids is less than 20% of the total interface between eyelet and pad. Void(s) do not visibly expose base metal. The presence of nonacceptability.

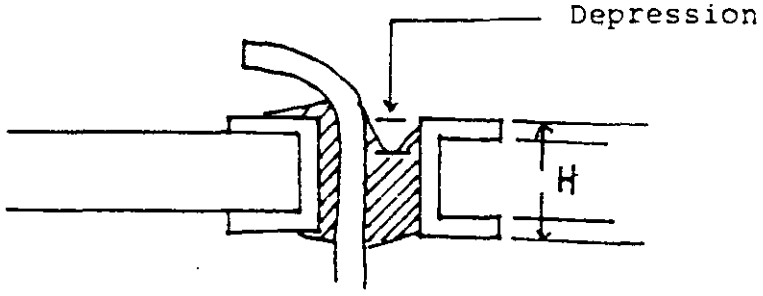


ACCEPTABLE

40.2.2 Maximum solder. A solder joint where it can be established that the lead exists and solder does not overflow beyond the confines of the connection area (the pad and tracks). Lead and pad must be well wetted.

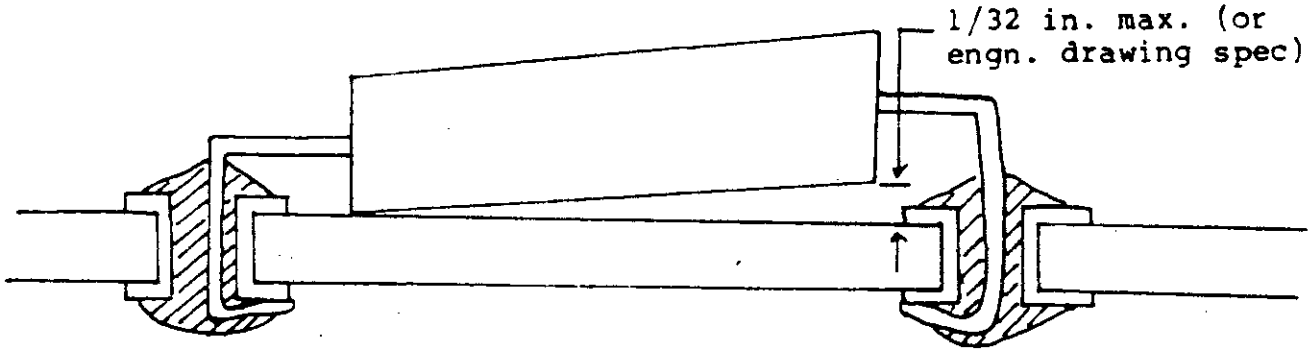
40.3 Lifting of pad. Lifted pads shall be controlled. Quantity of boards with lifted pads shall be controlled by the contractor as follows: Lifted pads exceeding 1/3 lifted to 0.40% AQL. Lifted pads less than 1/3 lifted to 1.0% AQL. Damage must not be quadrant enclosing track. Damage not to break into plated through hole. No damage to opposite hole.

40.4 Lead, solder side. Exposed end (except copper) are visible at the end of the end. Lead is not closer than .015% to adjoining conductive circuitry. Maximum solder is permissible. Solder fillet is depressed, but the depression is less than 25% of the board thickness.



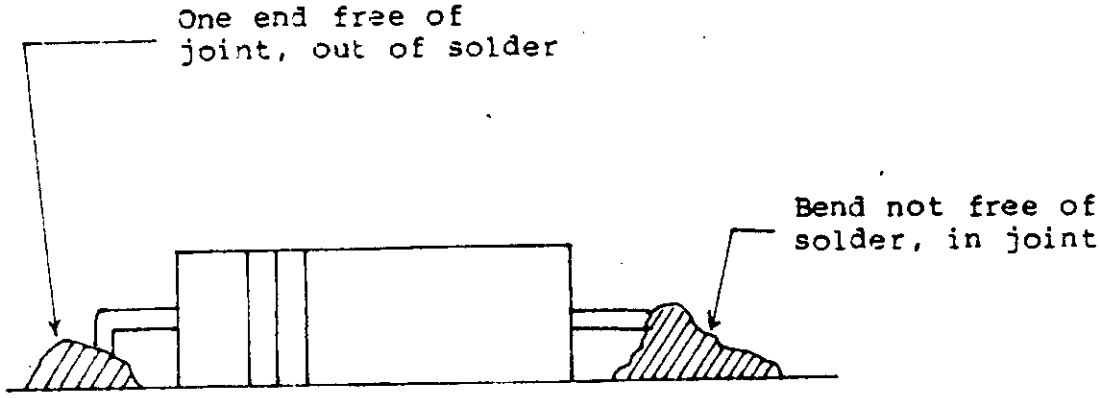
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40.5 Component alignment.

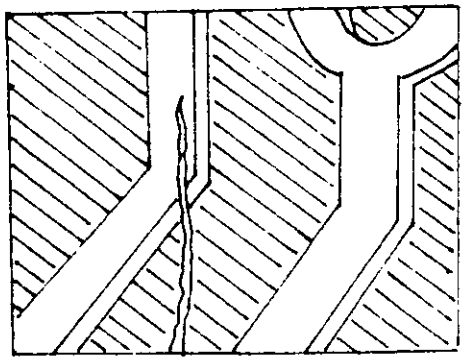


Vertical misalignment of component and printed wiring board is not in excess of 1/32 in. (or as specified on engineering drawing.)

40.6 Strain relief. For axial lead components having a 0.080 max. diameter body, allow one lead bend to be in solder joint, providing opposite lead bend is free of solder.



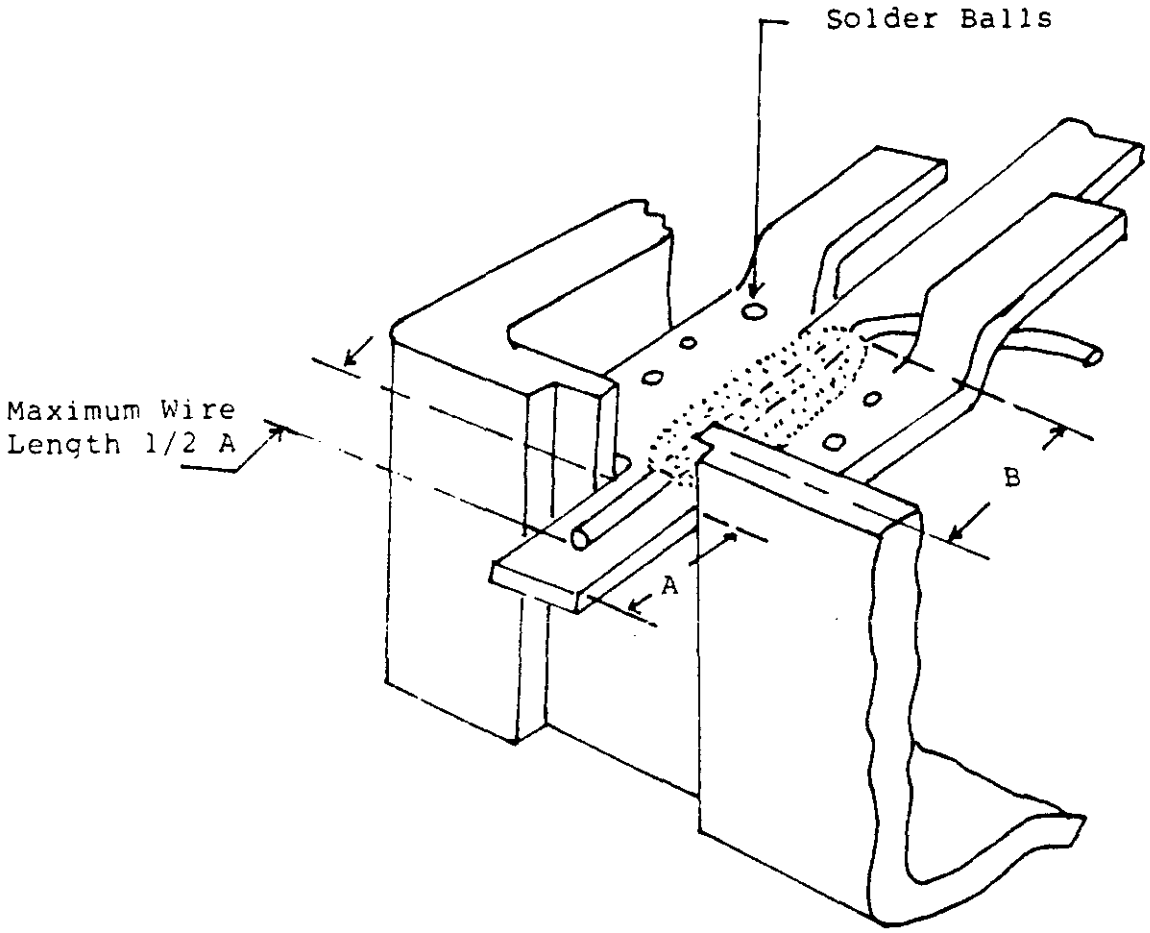
40.7 Scratches.



1. Scratch does not reduce cross-sectional area of conductor more than 20%
2. Does not visible expose base metal.

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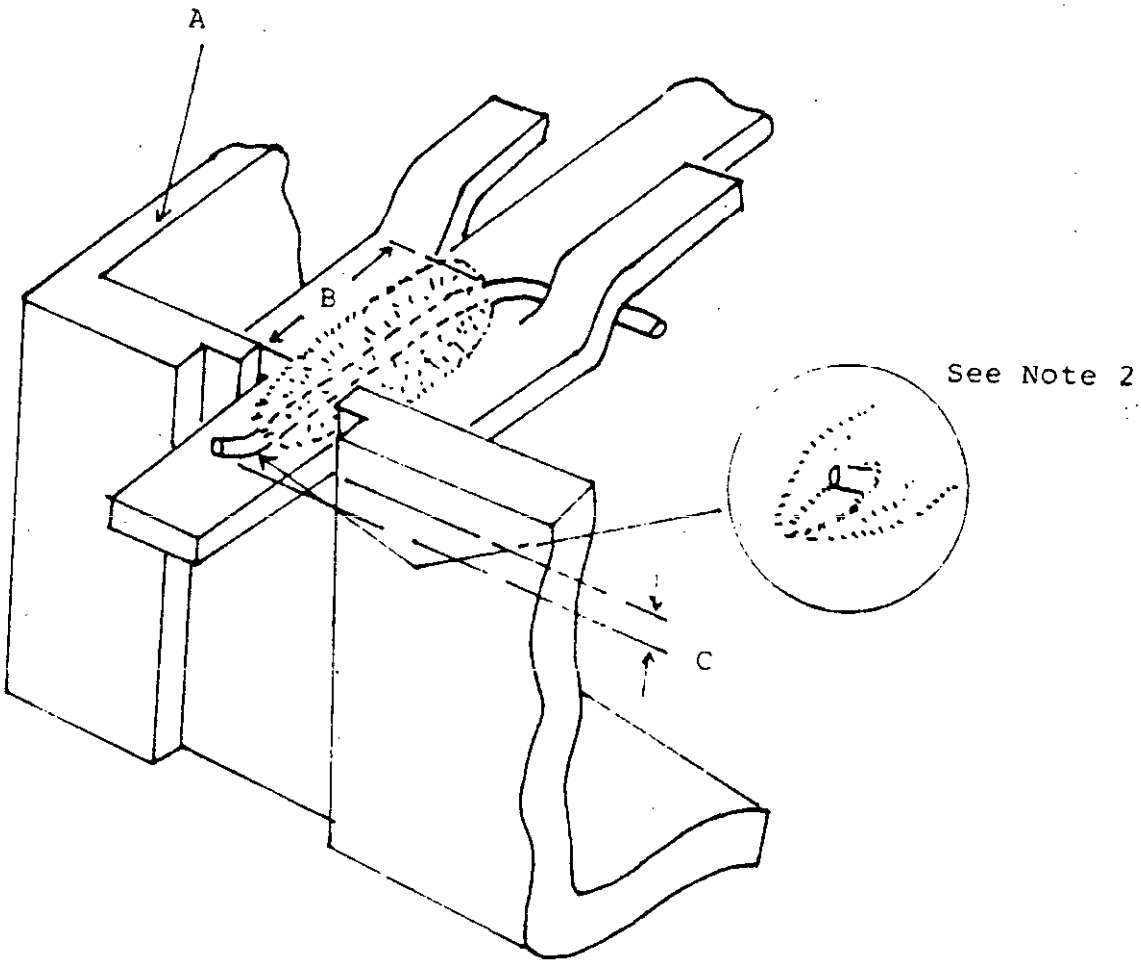
40.8 Magnetometer Soldering. This section describes the acceptance criteria for solder connections of magnet wire to terminal. All examinations shall be performed with 10x magnifications (microscope).



PREFERRED

1. Good wetting of wire and terminal over entire length "B".
2. Solder is free of holes.
3. Contour of the wire visible.
4. Wire is extended along the length of terminal.
5. Protrusion of wire is less than one-half of Terminal Length (A).
6. Solder balls are permissible in the potting area.
7. Bare copper end of wire is permissible.
8. Wire routed between terminal tabs, but no mechanical wrap is required.
9. Melted plastic may be present in the vicinity of solder joint (not allowed in solder joint).

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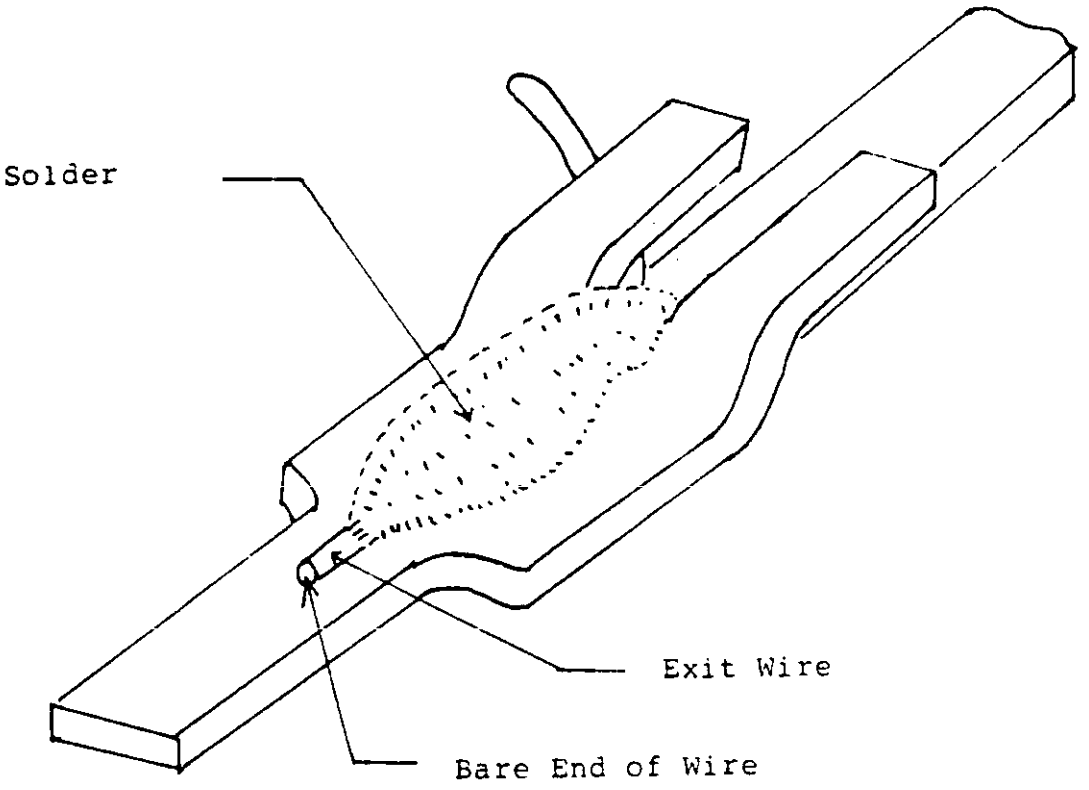
ACCEPTABLE PROTRUSION OF WIRE

1. Protrusion of the wire is above the surface of the terminal, but below the mold surface "A".

2. Imprint of wire after wire lifting during breakoff is permissible in area "C".

3. The lead must extend into area "C".

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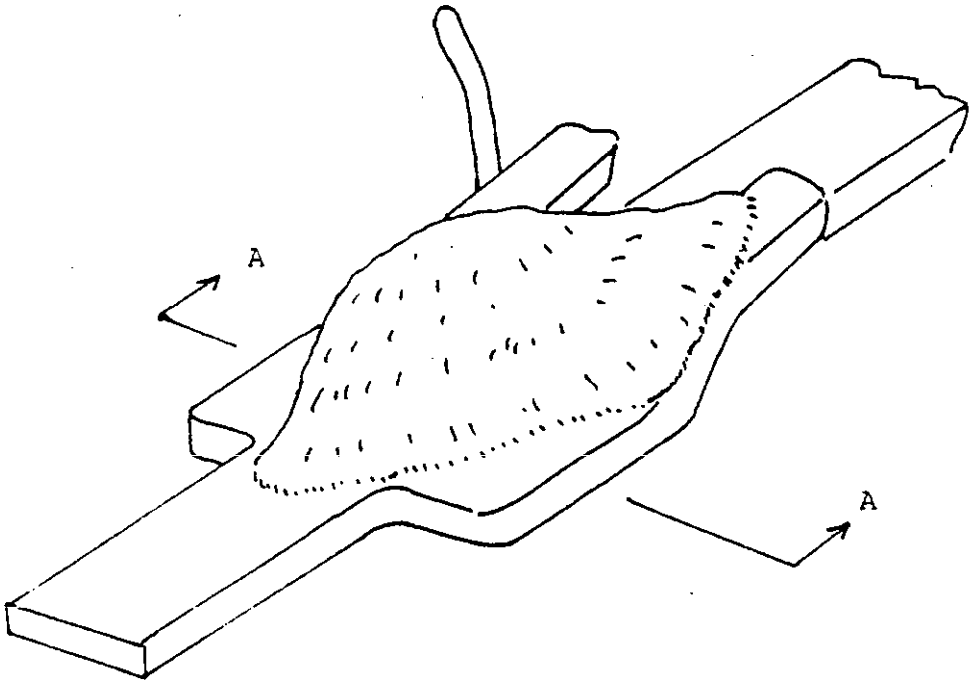
ACCEPTABLE

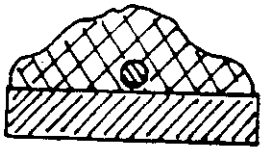
1. Bare end of wire is permissible.

2. Contour of wire not visible but wire ends are visible and show evidence of wetting entrance or exit.

3. Irregular surface; slightly convex fillet.

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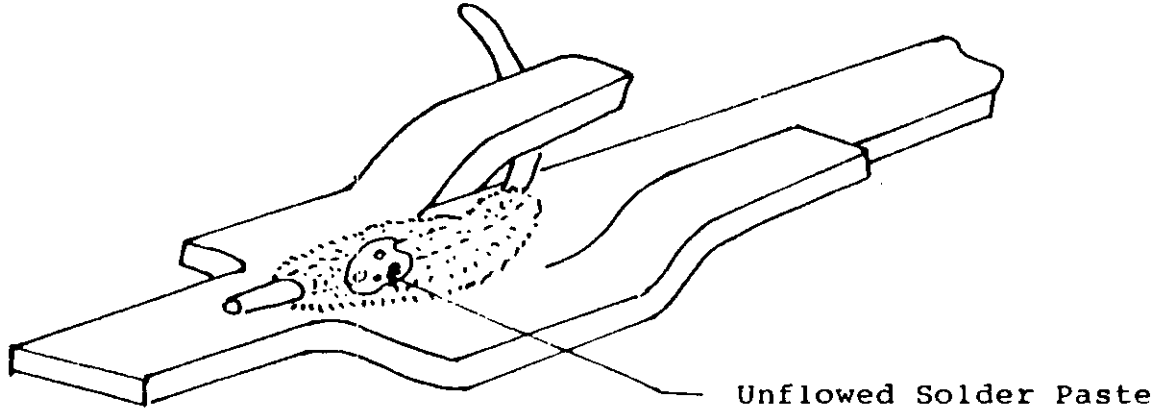
Section A-A

UNACCEPTABLE

1. The bare copper end of the wire is not visible.

2. Cannot be established that wire exists or that proper connection has been made.

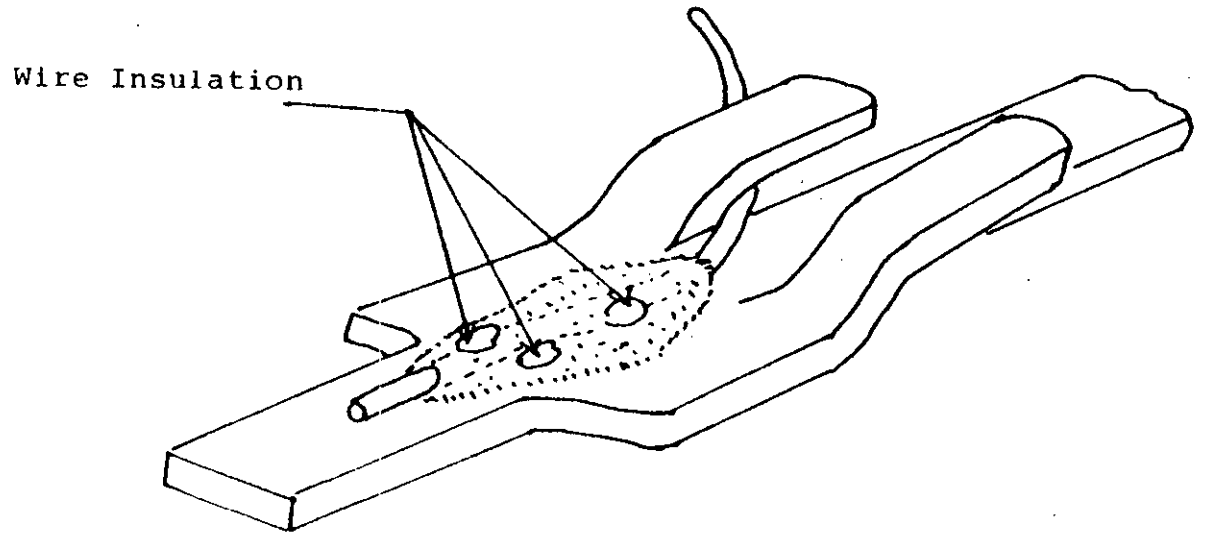
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UNACCEPTABLE

1. Evidence of solder past in solder connection.

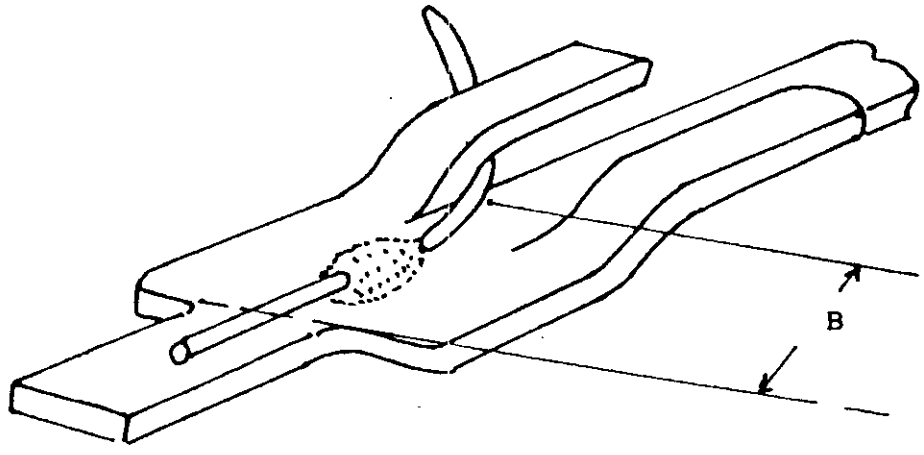
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UNACCEPTABLE

1. Evidence of contamination of solder connection.

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UNACCEPTABLE

1. Solder covers less than 50% of length "B".
2. Solder is not continuously wetted.
3. Voids exceeding 20% of solder joint present.

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50 PRINTED WIRING BOARD ASSEMBLIES

Following wave soldering, printed wiring board assemblies shall be inspected (under adequate lighting and magnification) for soldering workmanship.

50.1 Major defect inspection.

a. Inspection shall be to a 0.40 percent AQL, MIL-STD-105 Level II, normal sampling plan.

b. A board will be defined as defective during inspection if:

1. The bottom contains any joints defined as unacceptable by any of the documentation inf sections 20, 30 and 40 of this appendix.

2. The top contains any joint defined as unacceptable by the documentation listed in b.1 above except twelve (12) joints are allowed to be dewetted. A dewetted joint on the top must have evidence of solder in the hole when viewed from the top.

3. A maximum of thirty-five (35) joints with 0 to 20% voids will be allowed on each board; and no more than twenty (20) joints with 0-20% voids will be allowed on any one side of the printed wiring board.

50.2 Minor defect inspection.

a. Inspection shall be on the sample size as specified in paragraph 50.1a above.

b. A board will be defined as defective if it contains three (3) or more minor defects.

c. Accept/reject criteria for the lot shall be to a 4.0% AQL, MIL-STD-105 Level II, normal sampling plan.

d. Minor defects are defined as the following:

1. Conductive surfaces closer than .010" but greater than .005" assembly characteristics only, not relative to the printed wire board construction.

2. Copper showing:

a. Clipped lead - evidence of good wetting and reflow must be present.

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b. Non-clipped lead - characteristic of good solder joint, i.e., wetting, etc., must be present.

3. Solder splash (attached and not shorted).

4. Solder balls less than .005".

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-63241B	ELECTRONICS ASSEMBLY FOR NTMES, M710/M711 SYSTEM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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